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DECLARE vehicleType, category, permit, emergency: STRING
DECLARE valid: BOOLEAN
DECLARE zone, highestzone: CHAR
DECLARE spacesneeded, OccA, OccB, OccC: INTEGER
DECLARE remA, remB, remC, accept, reject: INTEGER
DECLARE
```

```
OUTPUT "Enter number of vehicles expected: "
INPUT N
FOR i ← 1 to N
    valid ← FALSE
    While valid = FALSE DO
        OUTPUT
        vehicleType ← INPUT "Enter vehicle type (C=Car, B=Bus, V=Van)"
        category ← INPUT "Enter user category (F=Faculty, S=Student, G=Visitor)"
        permit ← INPUT "Does the vehicle have a valid permit? (Y/N)"
        IF vehicleType = 'C' OR vehicleType = 'B' OR vehicleType = 'V'
        THEN
            IF category = 'F' OR category = 'S' OR category = 'G'
            THEN
                IF permit = 'Y' OR permit = 'N' THEN
                    valid ← TRUE
                ELSE OUTPUT "Invalid permit value. Enter again"
```

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ELSE OUTPUT "Invalid category value, Enter again"
ELSE OUTPUT "Invalid vehicle type, Enter again"
EndIf
EndWhile
IF permit = 'N' THEN
    INPUT emergency
    While emergency <= 'Y' AND emergency != 'N' DO
        OUTPUT "Invalid emergency value Enter Y or N"
        INPUT emergency
    ENDWhile
ELSE emergency ← 'N'
EndIf
IF vehicleType = 'V' THEN
    spacesneeded ← 2
ELSE
    spacesneeded ← 1
ENDIF
IF emergency = 'Y' THEN
    IF category = 'F' THEN
        IF OccA + spacesneeded <= 20 THEN
            OccA ← OccA + spacesneeded
            zone ← 'A'
            assigned ← TRUE
        ELSE
            OUTPUT "Rejected : no spaces available in zone"
        ENDIF
    ELSE IF category = 'S' THEN

```

Start

IF 'OccB + spacesneeded <= 40 THEN

OccB ← OccB + spacesneeded
Revenue ← p_B
Zone ← 'B'

assigned ← TRUE

ELSE

OUTPUT "No spaces available in Zone B"

ENDIF

ELSE IF category = 'G' THEN

IF OccC + spacesneeded <= 15 THEN

OccC ← OccC + spacesneeded

~~assigned~~ ← TRUE

ELSE

OUTPUT "No spaces available in Zone C"

ENDIF

ENDIF

PAC Chart Q5:

Problem	Process vehicles, validate their information, determine parki where to park and assign suitable zones
Analysis	Validate vehicle type, category, permit and emergency. Check eligibility, required parking spaces and remaining spaces. Apply zone rules where applicable.
Control	Outerloop for vehicles, validation if statements Decision for capacity and zones and vehicle type.
Output	Assigned Zone / rejection reason and parking summary.

IPO Chart Q5:

Input	Process	Output
Number of Vehicles	Process each vehicle	Assigned zone
Vehicle type	Validate input	Remaining zones
User Category	Check permit/emergency	Rejection reason
Parking permit	Determine eligibility	Total Processed
Emergency status.	Check remaining zones	Total accepted/rejected
	Assign correct zones	Cars/bikes/vans parked
	Update counters	Occupancy of Zones A, B, C
	Determine highest occupied ^{zone}	Remaining capacities
	Check whether campus is full	Highest Occupancy zone
		Campus full
		Campus full / not full

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